Linear independence

Let $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$; all coefficients are real.

Linearly independent: the only solution of $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = \mathbf{0}$ is $t_1 = \cdots = t_k = 0$ (the **trivial** solution).

Linearly dependent: the same equation has a **nontrivial** solution: some $t_i \neq 0$. Any list containing $\mathbf{0}$ is dependent. For three or more vectors, pairwise nonparallel does not imply independence.

A test of independence: form $M = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$ and row-reduce the homogeneous system $M\mathbf{c} = \mathbf{0}$. The vectors are independent exactly when every column is a pivot column.

Theorem: if the vectors are independent, each vector in $\operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ has a unique representation as a linear combination of them.

Adding a vector: if $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent and $\mathbf{v} \in \mathbb{R}^n$, adjoining $\mathbf{v}$ preserves independence if and only if $\mathbf{v} \notin \operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Basis and dimension

Let U be a subspace of $\mathbb{R}^n$.

Theorem: if U is spanned by m vectors and contains k linearly independent vectors, then $k \leq m$.

Basis: a set of vectors in U that is linearly independent and spans U.

Theorem: any two bases of U have the same number of vectors.

Dimension: for $U \neq \{\mathbf{0}\}$, $\dim(U)$ is the number of vectors in a basis. By convention, $\dim(\{\mathbf{0}\}) = 0$.

Existence and extension: if $U \neq \{\mathbf{0}\}$, then U has a basis and $\dim(U) \leq n$. Any independent set in U extends to a basis by adding vectors from any fixed basis of U. Any spanning set for U contains a basis.

Theorem: if $\dim(U) = d$ and B consists of exactly d vectors in U,

$$B \text{ independent} \iff B \text{ spans } U \iff B \text{ is a basis of } U.$$

Theorem: for subspaces $U \subseteq W \subseteq \mathbb{R}^n$, $\dim(U) \leq \dim(W)$; if their dimensions are equal, then $U = W$.

Ordered bases and coordinates

Ordered basis: $B = (\mathbf{b}_1, \dots, \mathbf{b}_d)$ is a basis of a d-dimensional subspace $U \subseteq \mathbb{R}^n$ with a specified order.

Coordinate vector: for $\mathbf{v} \in U$, the coefficients in $\mathbf{v} = t_1\mathbf{b}_1 + \cdots + t_d\mathbf{b}_d$ are unique, and

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_d \end{bmatrix} \in \mathbb{R}^d, \qquad C_B(\mathbf{b}_i) = \mathbf{e}_i \quad (1 \leq i \leq d).$$

Here $\mathbf{e}_i$ is the ith standard basis vector of $\mathbb{R}^d$; the order of B determines the order of coordinates.

Column space, row space, and rank

Let A be an $m \times n$ real matrix.

Column space: $\text{col}(A)$ is the span of the columns of A in $\mathbb{R}^m$; $\text{col}(A) = \text{im}(A)$.

Row space: $\text{row}(A)$ is the span of the rows of A in $\mathbb{R}^n$.

Lemma: row-equivalent matrices have the same row space. The nonzero rows of a row echelon matrix are linearly independent and form a basis of its row space.

Rank theorem and definition: $\text{rank}(A) = \dim(\text{col}(A)) = \dim(\text{row}(A))$.

Computing bases: if R is a row echelon form of A with pivots in columns $j_1, \dots, j_r$, its nonzero rows form a basis for $\text{row}(A)$; columns $j_1, \dots, j_r$ of the *original* matrix A form a basis for $\text{col}(A)$.

Warning: row operations need not preserve column space; use R to locate pivots, but take column-space basis vectors from A.

Rank identities: $\text{rank}(A) = r$ is the number of pivots in any row echelon form of A; $\text{rank}(A^T) = \text{rank}(A)$.

Nullity and rank-nullity

Nullity: $\text{nullity}(A) = \dim(\text{null}(A))$, where $\text{null}(A) = \{\mathbf{x} \in \mathbb{R}^n \,:\, A\mathbf{x} = \mathbf{0}\}$.

Rank-nullity theorem: if $\text{rank}(A) = r$, the $n - r$ basic solutions of $A\mathbf{x} = \mathbf{0}$ obtained by Gaussian elimination form a basis for $\text{null}(A)$. Thus

$$\dim(\text{im}(A)) = r, \qquad \text{nullity}(A) = n - r, \qquad \text{rank}(A) + \text{nullity}(A) = n.$$

Nullity counts free variables. With no free variables, $\text{null}(A) = \{\mathbf{0}\}$ and nullity is zero.

Null-space identity: if $A\mathbf{z} = \mathbf{0}$, then $A(\mathbf{x} + t\mathbf{z}) = A\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$ and $t \in \mathbb{R}$.